

Chemistry Ch 4.4: Characteristic Properties

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Property	Period mein	Group mein (neeche)
Metallic character	Ghatta hai	Barhta hai
Reactivity (Group 1)	not applicable	Barhti hai
Density	not applicable	Barhti hai

Metallic character ka matlab

Dhat hone ka matlab hai **electron denay ki aamadgi**. Group mein neeche atom **bara** hota hai, bairooni electron **door** hota hai, is liye **asaani se** nikal jata hai. Isi liye neeche jate hue dhat pan barhta hai. Aur period mein dayein jate hue atom chhota hota hai, electron mazbooti se pakra hota hai, is liye dhat pan ghatta hai.

Alkali metals (Group 1) ki khasoosiyat

Naram, kam kasafat wali, aur **bohat** reactive. Paani se mila kar hydrogen deti hain.

Neeche jate hue **reactivity barhti** hai: Li se Na, Na se K, aur K sab se zyada tez.

Peshangoi: agar Group 1 ke kisi naye element ke baare mein poochha jaye, to jo rujhan oopar hai wohi aage barha kar jawab likhein. Yehi kitab ka section 4.4.5 hai.

Anjaan element ki jagah dhoondna

Us ki **electronic configuration** likhein. Sab se bara shell = **period**. Bairooni electrons = **group**. Aakhri sub-shell = **block**. Phir us group ki khasoosiyat us par lagu kar dein.

Quiz

1. How do **metallic character**, **reactivity** and **density** change as you go down a group? Explain the reason for the metallic character trend.
2. Give three properties of the alkali metals, and explain how you would predict the behaviour of an unknown group 1 element placed below potassium.